QUALYXUS

Air-Gapped Edge AI for Visual Inspection & Process-Quality Intelligence

- **Startup Name:** Qualyxus
- **Founder & Lead Architect:** Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

> **We solve** late-stage scrap and escaped defects **by providing** an on-premise edge-AI platform that inspects every part and links each defect to the machine cycle that produced it, **to help** automotive, aerospace & defense, and industrial manufacturers **catch** process drift and machine wear before they become scrap or downtime, with all data staying inside the plant.

Compelling Problem Statement

The Problem: Quality and Machine Data Live in Separate Silos

Plants run two disconnected systems, so the defect and its cause are rarely seen together:

1. Machine Vision (Post-Mortem Sorting)

- Inspects parts **after** they are made, so scrap is caught only after material, machine hours and labour are spent.
- Rule-based setups need re-programming for every new part and struggle with variable cosmetic defects.
- Says pass or fail, but not **why**: blind to the machine conditions that caused the defect.

2. Machine Telemetry (Blind to Outcomes)

- Monitors temperatures, pressures, spindle load and vibration to prevent breakdowns.
- Never learns which parts came out bad, so it cannot tell which signal patterns cause defects.
- A machine can stay inside every alarm limit while slowly drifting into producing scrap.

Target Customer Profile: Plant Heads, VPs of Quality and Maintenance Heads running automated machining, moulding, casting, stamping, welding and assembly lines in automotive, aerospace & defense, and industrial manufacturing.

Solution Offered & Innovation Proposed

The Causal Bridge: Every Defect Linked to Its Machine Cycle

Qualyxus fuses camera inspection with machine-cycle data on one on-premise edge box, so quality and maintenance teams see the same cause.

Camera + Machine Data
Industrial cameras; Modbus TCP / OPC UA

On-Prem Edge Engine
Inspect, link to cycle, score quality & wear

Advisory PLC Signal
Reject / alert flag; plant logic decides

Core Deep-Tech Innovations

1 Learns from Good Parts

Anomaly detection (Dinomaly / PatchCore on DINOv2) trains on a few hundred good parts, with no defect library needed. AUROC 0.95–0.99 on 4 categories of the public VisA benchmark.

2 Calibrated False Rejects

Reject thresholds are set by statistical (conformal) calibration with a stated confidence bound, so the plant knows its false-reject rate before go-live.

3 Root Cause & Machine Health

Each part is linked to the cycle that made it. Cycle drift flags process problems and component wear (check rings, tools, dies) before they cause scrap, with approval-gated fixes (simulation-validated).

4 Air-Gapped by Design

Runs fully on-premise on an NVIDIA GPU; no cloud needed in production. Shadow mode, one-command rollback and a tamper-evident audit log built in.

Where We Play

One Platform Across Automated Manufacturing Processes

The vision model learns any part from good samples; the cycle engine reads any machine over Modbus TCP / OPC UA, and the same signals track the wear behind the defects.

Process	Defects the camera sees	Machine signals that explain them	Wear it tracks	Status
Injection Moulding	Short shots, flash, sink marks, burns, warpage	Injection & cavity pressure, melt / mould temperature, cushion	Check ring, heater bands, mould cooling	**Built end-to-end; validated in simulation**
CNC Machining	Burrs, chatter and tool marks, missing features	Spindle load, accelerometer vibration, tool counter	Cutting tools, spindle bearings	**Vision + OPC UA ready; MTConnect / FOCAS next**
Die Casting & Forging	Surface porosity, cold shuts, cracks, flash	Shot speed, die temperature, intensification pressure, press force	Dies, shot-cylinder seals	**Vision + cycle engine ready; template next**
Stamping & Press Work	Cracks, splits, wrinkles, burrs	Tonnage curve per stroke	Dies and punches	**Vision + cycle engine ready; template next**
Welding	Spatter, surface porosity, undercut, missing welds	Current, voltage, wire-feed speed	Electrode tips, wire feeder	**Vision + cycle engine ready; template next**
Assembly & Final Inspection	Missing, misaligned or wrong-variant parts	Press-fit force, torque curves	Press and nutrunner tooling	**Vision ready; multi-camera verdicts built**

Built & Measured

What the Platform Already Does

0.95–0.99

Anomaly-detection AUROC on 4 categories of the public VisA benchmark (previous baseline: 0.56–0.96).

1.8%

False-reject rate on held-out good parts with conformal-calibrated thresholds (synthetic parts).

100% → 6.7%

False rejects on loosely fixtured parts after automatic part registration, with defect recall unchanged (synthetic).

0.83

Mask IoU for defect segmentation (RF-DETR-Seg): 100% recall and defect-type accuracy, 16.5 ms per image on a laptop GPU (synthetic).

~4 s

To promote a new model live with no downtime; 98.5% shadow-mode agreement in closed-loop simulation.

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Audit records verified intact in the hash-chained, tamper-evident log; 75 automated tests in passing.

Results on the public VisA benchmark and on our synthetic part and process simulator. Measuring the same metrics on real lines is the goal of the paid pilots.

Market & Business Model

A Large Market, Priced Below One Inspector

$45B

TAM

Global AI visual inspection + predictive maintenance, 2025.[1]

$590M

SAM

~1.7 lakh inspection points across 6 processes, India + SE Asia × ~₹3L / yr.[2]

$5.5M

SOM

2,000 Indian lines × ₹2.4L / yr (~₹48 Cr ARR), 5-year target; exports additive.

Pricing & Customer Payback

- **Station:** ₹2–2.5L one-time per line (camera, lighting, Jetson, 1–2 day install).
- **Software:** ₹20,000 / month per line, less than one inspector, covering all 3 shifts.
- **Machine-health add-on:** from ₹5,000 / machine / month (pilot phase).
- **Paid 60-day pilot:** ₹1L, credited on conversion.

Per line (planning assumptions)	Value
Annual value delivered	₹6L – ₹15L (up to ₹25L)
Year-1 cost to customer	₹4.4L – ₹4.9L
Customer payback	4 – 10 months
Year-2+ return on licence	2.5× – 6×
Our software gross margin	80%+ (target)

[1] MarketsandMarkets, Market Research Future; IMARC. [2] Bottom-up estimate from IMTMA, ACMA, Foundry Informatics Centre, AIFI and OEM installed-base data. ₹87 = $1.

Competition

Where Qualyxus Fits

	Legacy Vision (Cognex / Keyence)	AI Vision (SwitchOn)	Heavy Automation (Siemens / Bosch)	QUALYXUS
Approach	Post-line sorting	Primarily visual inspection	Machine condition monitoring	**Inspection + early warning on drift & wear**
Data used	Rule-based image tools (+ add-on DL)	Deep-learning images	Machine telemetry	**Images linked to machine cycles**
Deployment	Weeks of integrator programming	Vendor-led setup per part	Multi-month projects	**1–2 day advisory install**
Budget	Quality only	Primarily quality	Maintenance only	**Quality + maintenance**
Typical cost	₹1.2–7.5L per camera + integration	Not public	₹50L+ CapEx	**₹2–2.5L + ₹20k / month**

Cognex costs from Indian dealer listings (In-Sight 2000–8000); indicative only.

Go-to-Market & Roadmap

Land, Prove, Expand, Scale

1. Land

Advisory install in 1–2 days; the plant's PLC keeps all stop authority.

2. Prove

60-day paid pilot in shadow mode beside manual inspectors.

3. Expand

5–8 lines per plant within 24 months; plants run 20–100+ machines.

4. Scale

Machine distributors and automation integrators bundle Qualyxus.

Built Today

Inspection with calibrated thresholds, segmentation, part-to-cycle linking, risk model, model ops, audit log, dashboard.

Next 6 Months

Jetson deployment, CNC connectors (MTConnect / FOCAS), accelerometer input, machine-health index.

12–18 Months

360° multi-camera enclosure, casting / stamping / welding templates, remaining-life estimates.

Projected on 3 target pilot lines: ~₹35L annual value vs. ₹7.2L licence fees (~5× return). Measured pilot results will replace this projection.

Team & Funding Path

Staged Funding, Each Step Unlocked by Proof

Founder & Core Architect

Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

- Built the full platform: anomaly detection, segmentation, part-to-cycle linking, risk models, model ops.
- 100% founder-owned; incorporation planned with dhaRti support.

Strategic Advisors

- **Dr. Rakesh Kumar:** VP R&D, Rossell Techsys (ex-Moog, ex-Indian Navy)
- **Shankar Lal:** Discipline Chief, Cyient (21 years at HAL)
- **Ramesh Rana:** Operations Lead, SIG SAUER

Now · dhaRti seed ₹15L · 9 months

3 pilot stations, measured plant results, first paying customer.

Next · Grants via dhaRti

Startup India Seed Fund / NIDHI-PRAYAS, up to ₹20L each.

Then · ₹2 Cr pre-seed · 18 months

40+ paid lines, ~₹1 Cr software ARR, CNC connectors and Jetson shipped.

3-year plan: 20 → 90 → 300 paid lines · software ARR ₹0.48 Cr → ₹2.1 Cr → ₹7.0 Cr (planning assumptions).

What I Need from dhaRti

A ₹15L Seed Grant + Five Kinds of Support

₹15 Lakh

Seed grant, milestone-based, over 9 months

Tranche 1 · ₹5L · Month 0

Test rig + first station + company setup → rig live, Jetson benchmarks (M3)

Tranche 2 · ₹5L · Month 3

Two more stations + intern → 3 pilot stations in plants (M6)

Tranche 3 · ₹5L · Month 6

Pilot operations → measured results, first paying customer (M9)

Minimum to start: ₹5L (tranche 1).

Lab space

Desk, test-rig bench and GPU compute time.

Pilot introductions

3–5 plants in the Hubballi–Dharwad–Belagavi cluster.

Mentors

Manufacturing quality, industrial sales and IP.

IP & company setup

Patent assessment, incorporation, DPIIT recognition.

Investor access

Introductions for the ₹2 Cr pre-seed once pilots are live.

In return: an IIT Dharwad deep-tech startup in Industry 4.0, with measured results from real plants within 9 months and monthly progress reports.

QUALYXUS

Edge AI that traces every defect to its cause

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